Computational Biology

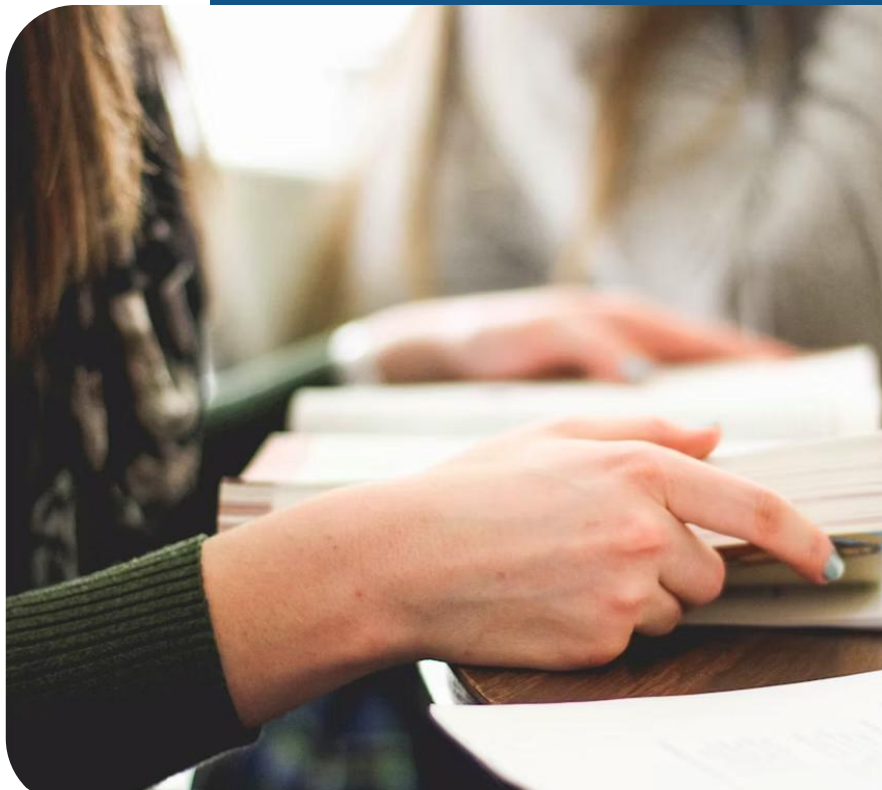
Session 7



Today's Agenda

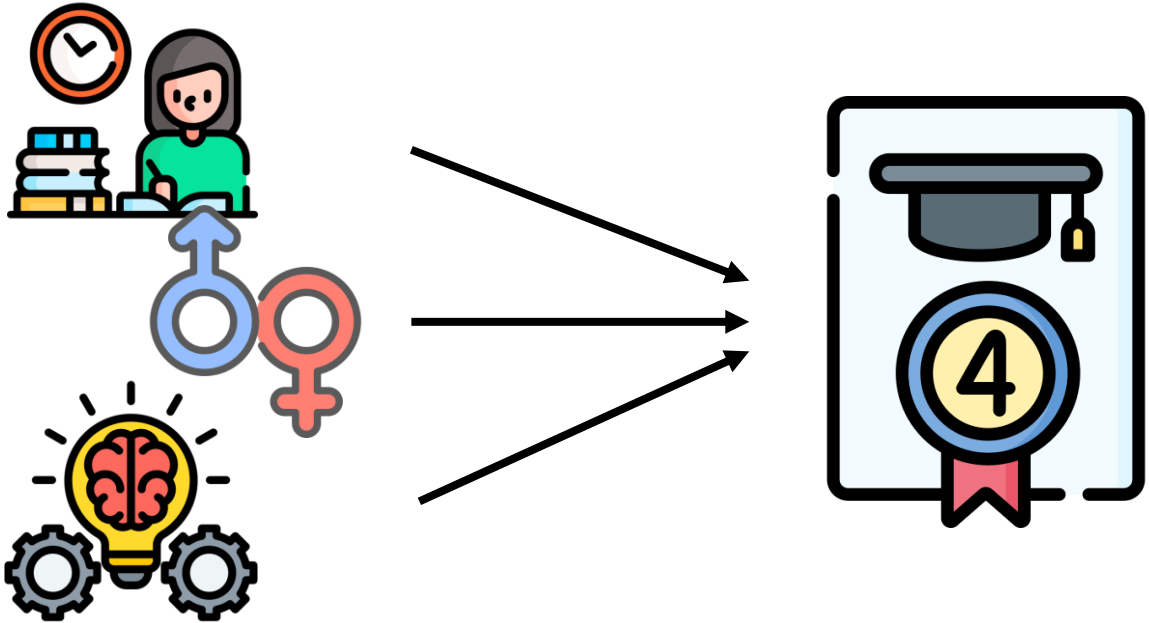
- Genome-Wide Association Study
- Polygenic Risk Score

Genome Wide Association Study (GWAS)



Let’s Review Regression

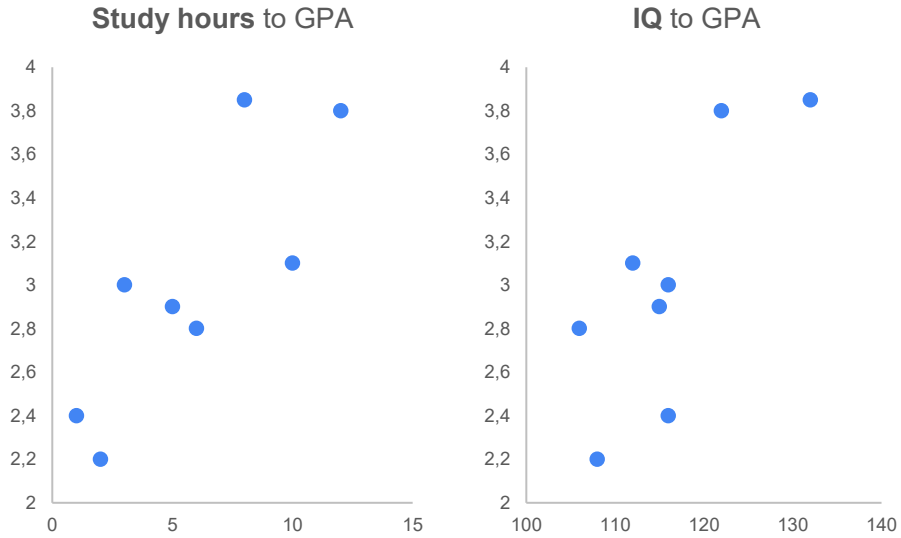
Let’s say we want to conduct an association study to examine whether weekly study hours and gender are associated with GPA, and to understand how these factors influence GPA.



Let’s Review Regression (1)

Data on these variables are collected from a sample of the student population. The study population is Binus University, Computer Science students.

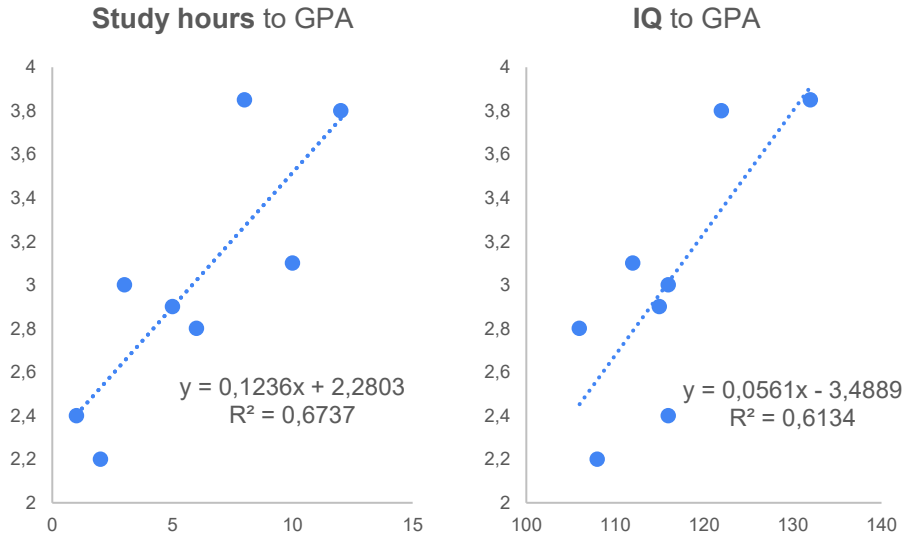
Name	Study hours	IQ	Gender	GPA
Andi	2	108	0	2.2
Ayu	1	116	1	2.4
Budi	5	115	0	2.9
Bella	3	116	1	3
Chandra	6	106	0	2.8
Citra	10	112	1	3.1
Dina	8	132	1	3.85
Dedi	12	122	0	3.8



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Name	Study hours	IQ	Gender	GPA
Andi	2	108	0	2.2
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Dedi	12	122	0	3.8



Let’s Review Regression (2)

Let’s do statistical analysis and modeling based on this data

```python
import pandas as pd
import statsmodels.api as sm

# Data
data = {
    "Study hours": [2, 1, 5, 3, 6, 10, 8, 12],
    "IQ": [108, 116, 115, 116, 106, 112, 132, 122],
    "Gender": [0, 1, 0, 1, 0, 1, 1, 0],
    "GPA": [2.2, 2.4, 2.9, 3.0, 2.8, 3.1, 3.85, 3.8]
}

df = pd.DataFrame(data)

df

# Define X and Y
X = df[["Study hours", "IQ", "Gender"]]
Y = df["GPA"]

# Add constant (intercept)
X = sm.add_constant(X)

# Fit model
model = sm.OLS(Y, X).fit()

# Show summary
print(model.summary())
```

OLS Regression Results

Dep. Variable:	GPA	R-squared:	0.935
Model:	OLS	Adj. R-squared:	0.887
Method:	Least Squares	F-statistic:	19.25
Date:	Sun, 12 Apr 2026	Prob (F-statistic):	0.00769
Time:	09:49:16	Log-Likelihood:	4.3751
No. Observations:	8	AIC:	-0.7502
Df Residuals:	4	BIC:	-0.4325
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-2.1770	1.214	-1.793	0.147	-5.548	1.194
Study hours	0.0914	0.022	4.209	0.014	0.031	0.152
IQ	0.0402	0.011	3.577	0.023	0.009	0.071
Gender	-0.0201	0.161	-0.125	0.906	-0.467	0.426

Omnibus:	1.176	Durbin-Watson:	1.284
Prob(Omnibus):	0.555	Jarque-Bera (JB):	0.779
Skew:	0.647	Prob(JB):	0.677
Kurtosis:	2.185	Cond. No.	2.02e+03

Let's Review Regression (3)

Let's review the result

	coef	std err	t	P>|t|	[0.025	0.975]
const	-2.1770	1.214	-1.793	0.147	-5.548	1.194
Study hours	0.0914	0.022	4.209	0.014	0.031	0.152
IQ	0.0402	0.011	3.577	0.023	0.009	0.071
Gender	-0.0201	0.161	-0.125	0.906	-0.467	0.426

Note:
The model is only valid for individual from the same population

$$GPA = -2.1170 + 0.0914 \times study\ hour + 0.0402 \times IQ - 0.0201 \times gender$$

In general association study, usually we used CL 95%, or p-value <0.05

P-value:

Study hour	: 0.014	(significant)
IQ	: 0.023	(significant)
Gender	: 0.906	(not significant)

Let's Review Regression (4)

Let's adjust to Logistic Regression (Binomial Prediction)

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \cdots$$

Name	Study hours	IQ	Gender	GPA	High Grade
Andi	2	108	0	2.2	0
Ayu	1	116	1	2.4	0
Budi	5	115	0	2.9	0
Bella	3	116	1	3	1
Chandra	6	106	0	2.8	0
Citra	10	112	1	3.1	1
Dina	8	132	1	3.85	1
Dedi	12	122	0	3.8	1

Let’s Review Regression (5)

Let’s do the logistic modeling

```
# Define X and Y (exclude Name & GPA)
X = df[["Study hours", "IQ", "Gender"]]
Y = df["High_GPA"]

# Add constant
X = sm.add_constant(X)

# Fit logistic regression
logit_model = sm.Logit(Y, X).fit()

# Show summary
print(logit_model.summary())
```

Coef:

const	≈	-16.10
Study hours	≈	1.48
IQ	≈	0.11
Gender	≈	0.35

Unfortunately, no variable is significant (95%):

- Study hour : 0.082
- IQ : 0.170
- Gender : 0.770

This may be due to very small data set
n=8

$$\log\left(\frac{p}{1-p}\right) = -16.10 + 1.48 \times Study\ hour + 0.11 \times IQ + 0.35 \times Gender$$

Let’s Review Regression (6)

Let’s add one more binomial predictor variable

Name	Study hours	IQ	Gender	Extra Course	GPA	High Grade
Andi	2	108	0	0	2.2	0
Ayu	1	116	1	0	2.4	0
Budi	5	115	0	1	2.9	0
Bella	3	116	1	0	3	1
Chandra	6	106	0	1	2.8	0
Citra	10	112	1	1	3.1	1
Dina	8	132	1	1	3.85	1
Dedi	12	122	0	1	3.8	1

Let's Review Regression (7)

Let's do the logistic modeling

```
# Define X and Y
X = df[["Study hours", "IQ", "Gender", "Extra_Course"]]
Y = df["High_GPA"]

X = sm.add_constant(X)

logit_model = sm.Logit(Y, X).fit()
print(logit_model.summary())
```

Coef:

const	≈	-18.5
Study hours	≈	1.35
IQ	≈	0.12
Gender	≈	0.25
Extra_Course	≈	1.80

Significancy level, let's use CL 90%:

- **Study hour** **: 0.09**
- IQ : 0.15
- Gender : 0.75
- **Extra course** **: 0.07**

$$\log\left(\frac{p}{1-p}\right) = -18.5 + 1.35 \times Study\ hour + 0.12 \times IQ + 0.25 \times Gender + 1.80 \times Extra\ Course$$

Let’s Review Regression (8)

Let’s do the prediction

Name	Study hours	IQ	Gender	Extra Course	High Grade	Prediction
Andi	2	108	0	0	0	0.001
Ayu	1	116	1	0	0	0.002
Budi	5	115	0	1	0	0.1
Bella	3	116	1	0	1	0.5
Chandra	6	106	0	1	0	0.25
Citra	10	112	1	1	1	0.97
Dina	8	132	1	1	1	0.999
Dedi	12	122	0	1	1	0.999
Rudi	2	100	0	1	?	0.05
Siti	12	130	1	0	?	0.98

Let’s Review Regression – Odd Ratio

Let’s do deep dive on the “Extra Course”

In logistic regression:

$$\text{Odds Ratio} = e^{b}$$

So, for **Extra Course**:
Coefficient = 1.80
Odds Ratio = $e^{1.80}$ = 6.05

This means:
Students who take the extra course have **~6 times higher odds** of getting a high GPA compared to those who don’t, holding other variables constant.

Odds = 1 → 50% probability
Odds = 6 → ~86% probability

Odd Ratio (1)

Let’s see other odd ratios:

```
const          ~ 0.00000001
Study hours    ~ 3.86
IQ             ~ 1.12
Gender         ~ 1.28
Extra_Course   ~ 6.05
```

Study hours OR = 3.86 → **each extra hour** multiplies odds ~4x
Extra course OR = 6.05 → bigger discrete jump

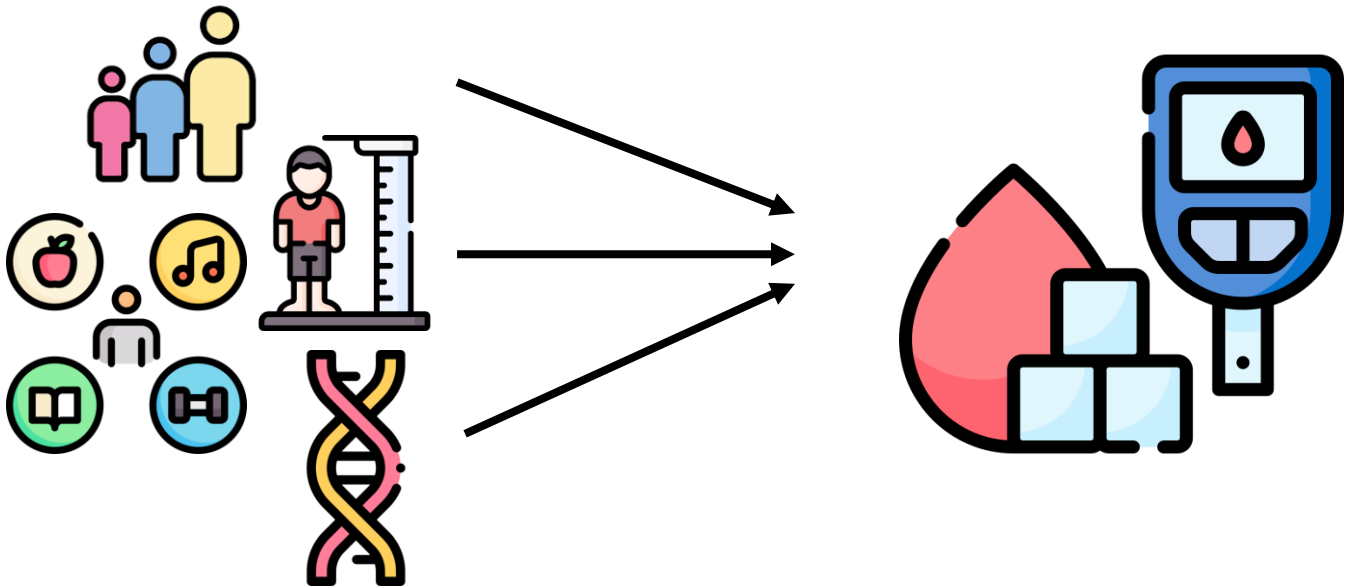
Let’s Review Regression

Let’s conclude

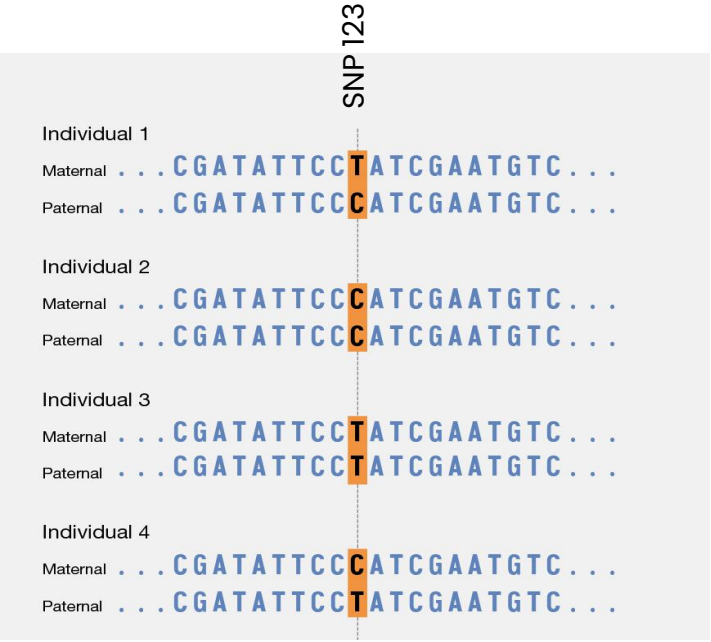
Taking an **extra course** was **significantly associated** with higher odds of achieving a high GPA (OR = 6.05, 90% CI: [x, y], $p < 0.1$), controlling for **study hours, IQ, and gender**.

GWAS

Now, let's apply the same approach to a disease or trait—specifically, diabetes. For the predictors, we use age, height, weight, gender, smoking status, and, importantly, genetic information.



Genetic Information Data



SNP 123:

T	(70%)
C	(30%) minor allele

Values range from 0 – 2. But genotype imputation may result in decimal values.

Homozygous major SNPs are marked as 0, homozygous minor as 2, and heterozygous as 1.

Indv	...	SNP 121	SNP 122	SNP 123	SNP 124	SNP 125	...
1		0	0	1	0	0	
2		0	0	2	0	0	
3		0	0	0	0	0	
4		0	0	1	0	0	

Merging Data

Indv	...	SNP 121	SNP 122	SNP 123	SNP 124	SNP 125	...	Age	Weight	Height	Gender	Smkg Status	Diabet
1		0	0	1	0	0		59	69	171	0	0	1
2		0	0	2	0	0		62	72	166	1	0	0
3		0	0	0	0	0		38	49	154	0	0	1
4		0	0	1	0	0		29	92	174	1	1	1
							...						
999		1	0	0	2	0		55	54	167	1	1	0
1000		0	1	1	0	0		34	51	155	0	0	0

Logistic Regression

Logistic Regression for each SNP with confounding variables

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \cdots$$

$$\text{SNP123:}\quad \log\left(\frac{p}{1-p}\right) = -3.5 + 0.75 \times SNP123 + 0.12 \times Age + 0.35 \times Weight + \ \ldots$$

P-value: 2×10^{-9}

Logistic Regression

. . .

SNP121: $\log\left(\frac{p}{1-p}\right) = -1.6 + 0.65 \times SNP121 + 0.11 \times Age + 0.32 \times Weight + \ldots$ P-value: 3×10^{-2}

SNP122: $\log\left(\frac{p}{1-p}\right) = -2.7 + 1.72 \times SNP122 + 0.32 \times Age + 0.31 \times Weight + \ldots$ P-value: 4×10^{-5}

SNP123: $\log\left(\frac{p}{1-p}\right) = -3.5 + 0.75 \times SNP123 + 0.12 \times Age + 0.35 \times Weight + \ldots$ P-value: 2×10^{-9}

SNP124: $\log\left(\frac{p}{1-p}\right) = -1.2 + 2.15 \times SNP124 + 0.42 \times Age + 0.42 \times Weight + \ldots$ P-value: 7×10^{-5}

SNP125: $\log\left(\frac{p}{1-p}\right) = -4.1 + 4.19 \times SNP125 + 0.15 \times Age + 0.22 \times Weight + \ldots$ P-value: 6×10^{-3}

. . .

GWAS Result

. . .

SNP121: P-value: 3×10^{-2}

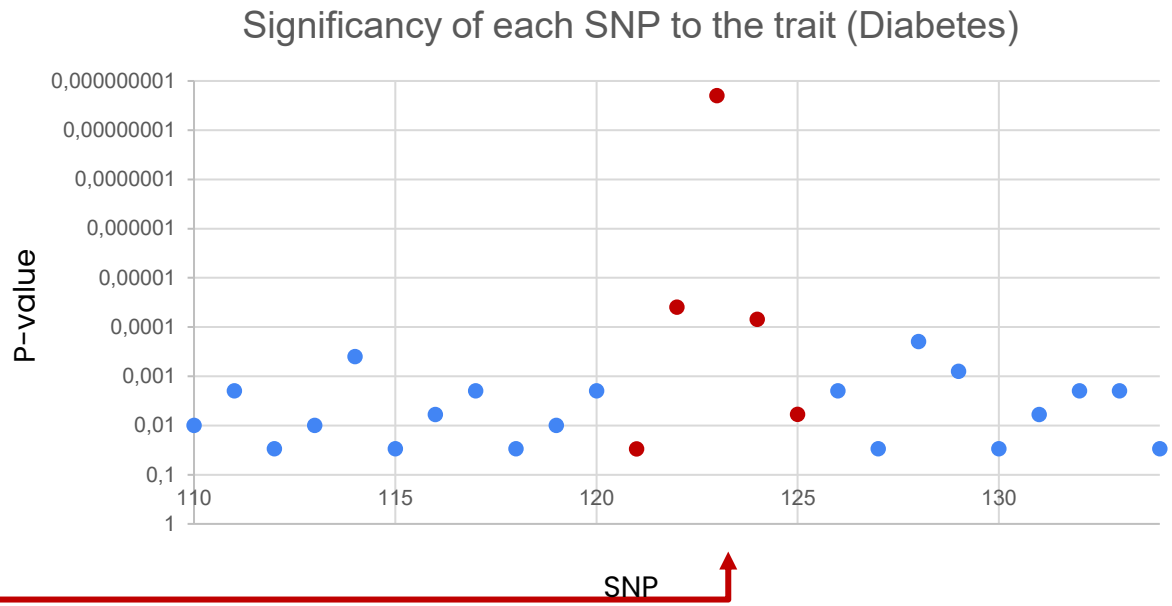
SNP122: P-value: 4×10^{-5}

SNP123: P-value: 2×10^{-9}

SNP124: P-value: 7×10^{-5}

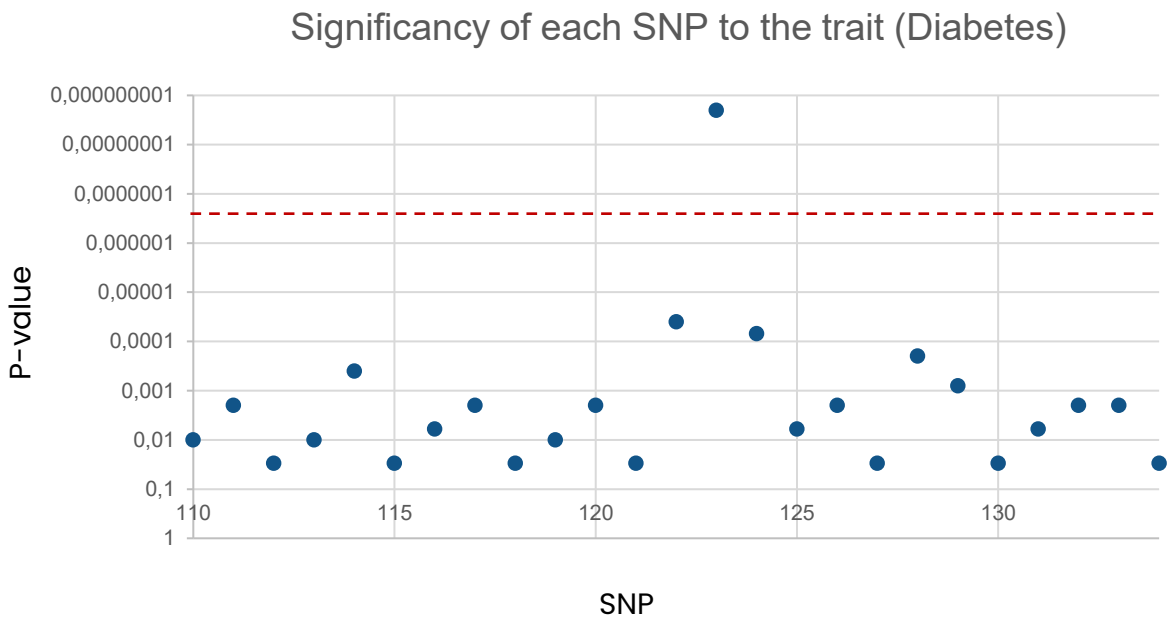
SNP125: P-value: 6×10^{-3}

. . .



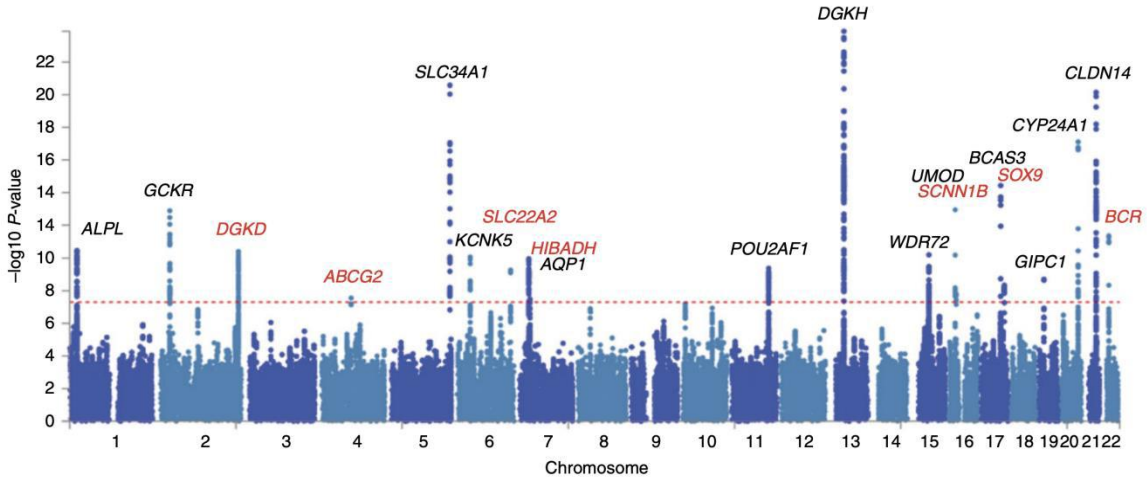
GWAS Result

The p-value threshold in GWAS analysis is **5e-8**.



GWAS Result

The p-value threshold in GWAS analysis is **5e-8**.



GWAS Result

In our example case, only SNP123 is significant. Hence let's look on its OR

In logistic regression:
Odds Ratio= e^b

So, for **SNP123**:
Coefficient = 0.75
Odds Ratio = $e^{0.75}$ = 2.13

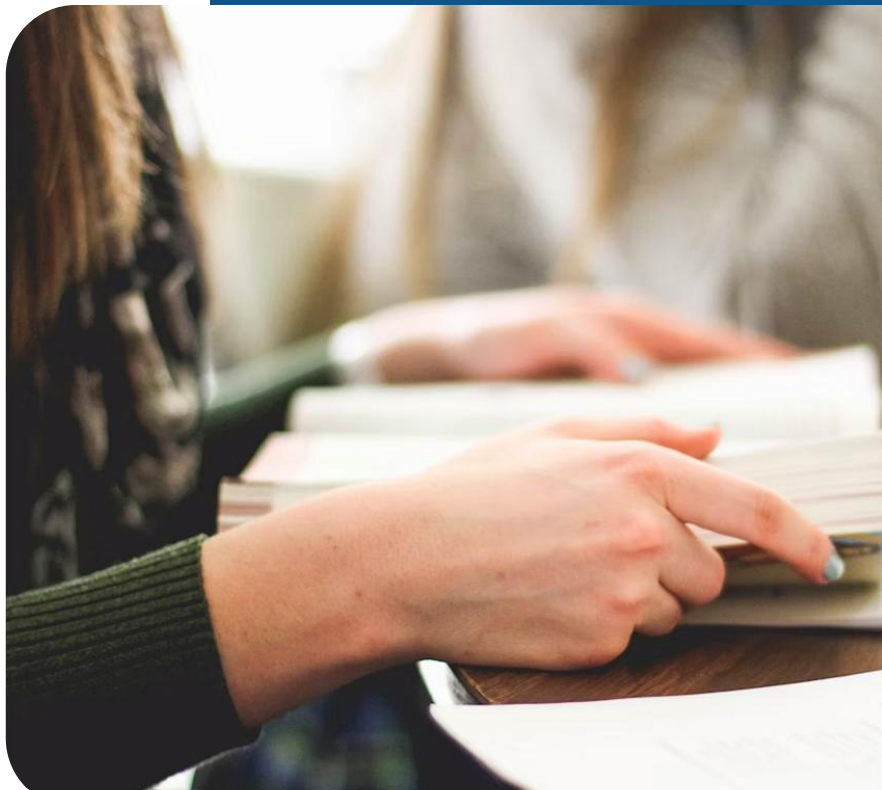
SNP 123:

T	(70%)
C	(30%) minor allele

This means:
Person who have minor allele in SNP123 **~2.13 times higher odds** of getting diabetes compared to those who don't, holding other variables constant.

Having a minor allele (C) in SNP123 was **significantly associated** with higher odds of having diabetes (OR = 2.13, p < 5e-8), controlling for **age, weight, height, gender and smoking status.**

Polygenic Risk Score (PRS)



Polygenic Risk Score (PRS)

A Polygenic Risk Score (PRS) is a way to estimate a person's genetic risk for a disease by combining the effects of many genetic variants (usually SNPs) across the genome. Most common diseases (like diabetes, heart disease) are not caused by a single gene, but by many genes each contributing a small effect.

"If we add up all these tiny genetic effects, how at-risk is this person?"

$$PRS = \sum_{i=1}^{N} b_i \times Gi$$

N = number of SNPs used
b_i= effect size of SNP_i (from GWAS)
G_i= genotype (0, 1, or 2 copies of risk allele)

PRS Example

Significant SNPs from across studies:

SNP	Risk allele	Effect size
SNP1	A	0.20
SNP2	G	0.35
SNP3	T	-0.15
SNP4	C	0.60

Genotype of one person:

SNP	Genotype	Risk alleles
SNP1	AA	2
SNP2	AG	1
SNP3	CC	2
SNP4	CT	1

$$PRS = \sum_{i=1}^{N} b_i \times Gi$$

$$PRS = (0.20 \times 2) + (0.35 \times 1) + (-0.15 \times 2) + (0.60 \times 1)$$

$$PRS = 1.4$$
$$OR = e^{1.4} = 4.06$$

This person has about **4.06 times the odds** of disease compared with a reference person with PRS = 0.

In PRS, not all SNPs increase risk. Some SNPs have negative effect sizes, meaning they are protective and reduce the overall score.

THANK YOU!